MILITARY SPECIFICATION

CYLINDERS, OXYGEN, STEEL, NON-SHATTERABLE, WELDED, 2100 PSI

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for 2100 psi (14,500 kPa) non-shatterable steel cylinders of welded construction.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

BB-A-1034	-	Air, Compressed, for Breathing Purposes.
BB-N-411	-	Nitrogen, Technical.
TT-C-490	-	Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Systems Engineering and Standardization Department (Code 53), Naval Air Engineering Center, Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8120

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TT-E-489	-	Enamel, Alkyd, Gloss (for Exterior and Interior Surfaces).
TT-L-32	-	Lacquer, Cellulose Nitrate, Gloss, for Aircraft Use.
TT-P-1757	-	Primer Coating, Zinc Chromate, Low Moisture Sensitivity.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-B-640	-	Boxes, Fiberboard, Corrugated, Triple-wall.

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MIL-P-116	-	Preservation, Methods of.
MIL-C-5501	-	Cap and Plug, Protective, Dust and Moisture Seal.
MIL-W-8611	-	Welding, Metal Arc and Gas, Steels, and Corrosion and Heat Resistant Alloys, Process for.
MIL-O-27210	-	Oxygen, Aviator's Breathing, Liquid and Gas.
MIL-T-27730	-	Tape, Antiseize, Tetrafluoroethylene, with Dispenser.
MIL-M-43719	-	Marking Materials and Markers, Adhesive, Elastomeric, Pigmented, General Specification for.
MIL-C-81302	-	Cleaning Compound, Solvent, Trichlorotrifluoroethane.

STANDARDS

FEDERAL

FED-STD-595	-	Colors.

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection By Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of U.S. Military Property.
MIL-STD-453	-	Inspection, Radiographic.
MIL-STD-810	-	Environmental and Test Methods.
MIL-STD-1264	-	Radiographic Inspection for Soundness of Welds in Steel by Comparison to Graded ASTM E 390 Reference Radiographs.
MIL-STD-1595	-	Aerospace Welder Performance Qualification.

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2.1.2 Drawings and publications. The following drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND

1684AS100 - Cylinder, Oxygen Nonshatterable, Welded (1,800 psi).

PUBLICATIONS

NAVAL AIR SYSTEMS COMMAND

SD-6 - Provisions Governing Qualification.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

COMPRESSED GAS ASSOCIATION (CGA)

Pamphlet No. C-1 - Methods for Hydrostatic Testing of Compressed Gas Cylinders.

(Application for copies of CGA pamphlets should be addressed to the Compressed Gas Association, Inc., Crystal Gateway 1, Suite 501, 1235 Jefferson Davis Highway, Arlington, VA 22202.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 8 - Tension Testing of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The cylinders furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3).

3.2 First article. When specified in the contract or purchase order (see 6.2.1h), a sample shall be subjected to first article inspection (see 4.4 and 6.4).

3.3 Materials. Materials shall conform to applicable specifications and shall be as specified herein and on applicable drawings. Materials which are not covered by specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practical weight, and suitable for the purpose intended.

3.3.1 Steel. The composition of the steel used in the construction of the cylinder shall consist of the required elements to provide cylinder properties and behavior as required by this specification.

3.4 Design and construction. The design and construction of the cylinders shall be in accordance with the applicable part number of Drawing 1684AS100 as specified in the contract or order (see 6.2.1c) and shall be of welded construction. The cylinders shall have been subjected to the initial permanent volumetric expansion test within 18 months prior to delivery.

3.4.1 Wire wrapping. Wire wrapping shall not be used in the construction of cylinders furnished under this specification.

3.4.2 Welding. Welding shall be performed in accordance with MIL-W-8611 and by a welder qualified in accordance with MIL-STD-1595. Welded areas shall be subjected to radiographic inspection.

3.4.3 Neck. Shrink rings or supporting collars shall not be used for strengthening the neck of the cylinder.

3.4.4 Closure. Unless otherwise specified herein, the threaded port of each cylinder shall be closed by a metal, removable, threaded plug or cap conforming to MIL-C-5501 or a commercially equal metal plug or cap, to protect the threads and prevent entrance of foreign matter or moisture.

3.5 Performance.

3.5.1 Visual examination and dimensional check.

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3.5.1.1 Visual examination. When visually examined as specified in 4.7.1.1, the cylinders shall conform to the requirements of this specification. Table IV, Classification of Defects for Visual Examination, shall be used to classify and enumerate the defects.

3.5.1.2 Dimensional check. When dimensionally checked as specified in 4.7.1.2, the cylinders shall conform to the requirements of Drawing 1684AS100. Any measurement deviating from the specified dimensions and tolerances shall be classified as a dimensional defect.

3.5.2 Cleanliness.

3.5.2.1 Non-volatile residue. When tested as specified in 4.7.2.1, the residue from the solvent used for the rinse shall be not greater than that of the clean solvent by more than .0015 grams per square foot of part surface area.

3.5.2.2 Particulate test. When tested as specified in 4.7.2.2, the solvent rinse shall not contain more than 5 particles between 100 and 175 micrometers per 100 ML of solvent; and no particles larger than 175 micrometers per 100 ML of solvent.

3.5.2.3 Halogen test. When tested as specified in 4.7.2.3, there shall be no evidence of cleaning compound residue.

3.5.3 Leakage. When tested as specified in 4.7.3, the cylinder shall not show any evidence of gas bubbles after 2 minutes of submergence.

3.5.4 Permanent volumetric expansion. When tested as specified in 4.7.4, the cylinder shall not have a permanent volumetric expansion greater than 10 percent of the total volumetric expansion.

3.5.5 Burst pressure. When tested as specified in 4.7.5, the cylinder shall not burst at a pressure less than 5,250 psi (36,199 kPa). Failure of cylinders shall occur in a ductile manner.

3.5.6 Endurance. When tested as specified in 4.7.6, the cylinder shall not show any evidence of material failure.

3.5.7 Flattening. When tested as specified in 4.7.7, the cylinder shall show no evidence of cracking when examined visually without any magnification.

3.5.8 Macrostructure defects. When tested as specified in 4.7.8, the cylinder shall show no evidence of abnormal segregation, pipes, cracks, seams or abnormal change in structure from the surface to the center.

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3.5.9 Radiographic inspection. When tested as specified in 4.7.9, welded sections of the cylinder shall be in accordance with MIL-STD-1264.

3.5.10 Fragmentation resistance.

3.5.10.1 Procedure I. When tested as specified in 4.7.11.1, the cylinder shall remain in one piece.

3.5.10.2 Procedure II. When tested as specified in 4.7.10.2, the cylinder shall be allowed to separate into two pieces. The projected area of any piece shall not be less than 2.0 square inches. The cylinder shall exhibit no evidence of shattering with small irregular pieces or splinters.

3.5.11 Physical properties. When tested as specified in 4.7.11, the samples shall not have a tensile strength greater than 200,000 psi (1.38×10^6 kPa) and elongation shall be not less than 6 percent.

3.6 Weight. The weight of the cylinder shall be not greater than the values specified on Drawing 1684AS100.

3.7 Color. The color of the cylinder shall be Green 14187 in accordance with FED-STD-595.

3.8 Finish. The external and internal surface shall be cleaned in accordance with Method I or VI of TT-C-490. The internal surface of cylinders fabricated of material other than corrosion-resistant steel shall be treated in accordance with Type I of TT-C-490. The external surface of all cylinders shall be painted with one coat of primer conforming to TT-P-1757 prior to application of the finish paint. The finish paint shall be two coats of lacquer in accordance with TT-L-32 or two coats of enamel in accordance with TT-E-489. Color of the paint shall be as specified (see 3.7). Extreme caution shall be exercised to prevent contamination during painting.

3.9 Identification marking. The cylinders shall be marked for identification in accordance with MIL-STD-130.

3.9.1 Standard markings. The cylinders shall be permanently and legibly marked using a pressure sensitive decal located as shown on Drawing 1684AS100. The adhesive on the decal shall conform to MIL-M-43719, Type I, Class 1. The markings shall be, not greater than 1/8 inch high and shall consist of the following (listed sequence mandatory):

a. Manufacturer's Name and CAGE (Commercial and Government Entity) code.

b. Government Part Number.

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c. 2100 PSI - 14,500 kPa.

d. Serial Number.

e. Date of Manufacture (month and year).

f. Date of permanent volumetric expansion test (month and year) with identifying symbol between the month and year.

g. A blank area for the addition of subsequent PVE test dates.

3.9.2 Additional markings. The following additional markings shall be applied by stenciling or clear pressure sensitive decal in two places diametrically opposite of each other (see Drawing 1684AS100 for location):

Aviator's Breathing Oxygen
2100 PSI - 14,500 kPa

The size of the markings and the space between each row of markings shall be not less than .12 times the outer diameter of the cylinder. The adhesive on the decal shall conform to MIL-M-43719, Type I, Class 1.

3.10 Workmanship. The cylinders shall be uniform in quality and shall be free from irregularities or defects which could adversely affect safety, performance, reliability or durability. The cylinders shall be free of oil, grease, fuel, water, dust, dirt or any other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3).

b. First article inspection (see 4.4).

c. Quality conformance inspection (see 4.5).

4.3 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in Table I.

4.3.1 Qualification samples. Qualification samples shall consist of the following:

a. Nine cylinders of the type (Part Number) for which qualification testing has been authorized.

b. Four material samples in accordance with ASTM E 8, cut from 4 completed cylinders.

c. Two sets of manufacturer's drawings in accordance with SD-6.

d. When applicable, certification from the original manufacturer that the distributor requesting qualification status is authorized to distribute the item, or to rebrand and distribute the item, as applicable.

Samples shall be forwarded to a test facility set forth in the letter of authorization to submit samples (see 6.3). The samples shall be plainly identified by securely attached durable tags marked with the following information:

Sample submitted by (name) (date) for qualification inspection in accordance with the requirements of MIL-C-29576 and number under authorization (reference authorizing letter and number, see 6.3).

4.3.2 Retention of Qualified Products Listing (QPL). The retention of qualification listings shall consist of verification every two years to determine compliance of the listed item with the requirements of this specification. Verification shall be by manufacturer's certification unless otherwise specified by the activity responsible for the QPL.

4.4 First article inspection. First article inspection shall consist of the examinations and tests specified in Table II, Listed Sequence Mandatory. If there is no Qualified Products Listing for the part number being procured, first article testing shall consist of the tests specified in Table I.

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4.4.1 First article samples. Unless otherwise specified herein, as soon as practicable after award of a contract or order, the manufacturer shall submit four cylinders of each part number specified in the contract or order. The samples shall be representative of the construction, workmanship, components, and materials to be used during production. When a manufacturer is in continous production of these cylinders from contract to contract, submission of further first article samples on the new contract may be waived at the discretion of the acquiring activity (see 6.2.1g). Approval of the first article samples or the waiving of the first articles inspection does not exclude the requirements of submitting to the quality conformance inspection. The first article samples shall be furnished to the Government as specified in the contract (see 6.2.1h).

4.4.1.1 First article information. Upon completion of the first article inspection program, the Government activity responsible for conducting the inspection program (see 6.2.1h) shall report to the contracting officer the results of the inspection program, with appropriate recommendations.

4.4.1.2 First article sample disposition. The samples will be destroyed during the inspection program and will not be returned to the contractor.

4.5 Quality conformance inspection. Quality conformance inspection shall consist of the examinations and tests specified in Table III (Listed Sequence Mandatory). The sampling and inspection levels and acceptance criteria shall conform to MIL-STD-105.

4.5.1 Sampling.

4.5.1.1 Inspection lot.

4.5.1.1.1 Cylinder. An inspection lot size shall be expressed in units of one cylinder of the same size, made essentially under the same conditions and from the same materials and components. The sample unit shall be one cylinder.

4.5.1.1.2 Packaging. An inspection lot size shall be expressed in units of one fully prepared shipping container, containing cylinders fully prepared for delivery, made from essentially the same materials and components. The sample unit shall be one shipping container, containing cylinders fully prepared for delivery with the exception that it need not be sealed.

4.5.1.2 Sampling for tests and examinations of cylinders. The sample size, acceptance criteria, tests, and examinations required for the cylinders shall be as specified in Table III.

4.6 Test conditions.

4.6.1 Gas. Unless otherwise specified herein, the gas used in testing the cylinders shall be either oxygen conforming to MIL-O-27210, Type I, or nitrogen conforming to BB-N-411, Type I, Class I, Grade B, or air conforming to BB-A-1034.

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4.6.2 Temperature and pressure. Unless otherwise specified herein, tests shall be conducted at local ambient temperature and barometric pressure. Test instruments shall be calibrated or adjusted according to their required usage in conducting individual tests. The temperature and barometric pressure shall be recorded at the time of inspection and, when required, the test results shall be corrected to normal temperature and pressure (NTP) conditions. NTP conditions are 70 degrees F (21.1 degrees C) and 29.92 inches of mercury (101.3 kPa).

4.6.3 Antiseize tape. To prevent damage to the cylinder threads or valves, male pipe threads of fittings installed in cylinders during tests shall be properly coated with antiseize tape conforming to MIL-T-27730 before being installed in the cylinder. The tape shall be applied according to instructions specified in MIL-T-27730. Tape shall be prevented from entering the inside of the fittings.

4.7 Inspection methods.

4.7.1 Visual examination and Dimensional Check.

4.7.1.1 Visual examination. Every cylinder shall be visually examined for conformance to the requirements of 3.5.1.1

4.7.1.2 Dimensional check. The cylinders shall be dimensionally checked for conformance to the requirements of 3.5.1.2.

4.7.1.3 Packaging. Each of the fully prepared shipping containers containing cylinders selected as a sample unit from the lot shall be examined to determine that the packaging, packing, and marking conform to Section 5 of this specification. The classification of defects, Table V, shall be used to classify the defects found.

4.7.2 Cleanliness. The cleanliness test shall consist of the non-volatile residue test (see 4.7.2.1), the particulate test (see 4.7.2.2), and the halogen test (see 4.7.2.3).

4.7.2.1 Non-volatile residue test. To perform this test, place a clean plug in the outlet of the cylinder and clean the surface around the plug and cylinder outlet with cleaning compound solvent conforming to MIL-C-81302, Type I. Wipe dry with a clean, lint free cloth. Calculate the internal surface area of the cylinder. Using the internal surface area of the cylinder and a ratio of 200 ML per sq. ft. (2.15 L/M^2), calculate the required amount of solvent (MIL-C-81302, Type I) to be used. In no case shall less than 150 ML of solvent be used. Remove the plug from the cylinder and pour in the solvent. Install the plug in the cylinder to seal the outlet. Agitate the cylinder for not less than ten minutes, in a manner to wash the entire internal surface. Immediately after completing the internal wash cycle, remove the plug and pour the solvent from the cylinder into a beaker. One-half of the solvent used for the rinse shall be used for determining weight of residue and the other half of the solvent shall be used for determining particle content (see 4.7.2.2). The weight of the residue shall be determined by separately evaporating to dryness one-half of the solvent used for the rinse and an equal amount of clean solvent. When tested in this manner, the cylinders shall conform to the requirements of 3.5.2.1. Upon successful completion of this test, the remaining one-half of the solvent shall be subjected to the test specified in 4.7.2.2.

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4.7.2.2 Particulate test. Upon successful completion of the test specified in 4.7.2.1, examine the remaining one-half of the solvent used for the rinse of particles. To determine the particle content, filter the solvent through a 1.2 micro milipore, or equivalent, membrane filter and inspect the filter surface for conformance to the requirements of 3.5.2.2. Upon successful completion of this test, the cylinders shall be subjected to the test specified in 4.7.2.3.

4.7.2.3 Halogen test. Upon successful completion of the test specified in 4.7.2.2, the cylinders shall be tested to detect any solvent residue. A Halogen or Halide detector, having a minimum sensitivity of not less than 3×10^{-4} standard cc's per second, shall be used for this test. The cylinders shall conform to the requirements of 3.5.2.3.

4.7.3 Leakage. The cylinder shall be charged to rated service pressure, 2100 +50/-0 psig (14,500 +345/-0 kPa). With this pressure maintained, the cylinder shall be submerged in water for 2 minutes. All parts of the cylinder shall be covered with not less than 1 inch (25.4 mm) of water. The cylinder shall pass the requirements specified in 3.5.3.

4.7.4 Permanent volumetric expansion. Permanent volumetric expansion of the cylinder shall be determined as specified in the hydrostatic test of Pamphlet C-1 of the Compressed Gas Association. The cylinder shall pass the requirements specified in 3.5.4. The cylinder shall have passed this test within 18 months prior to delivery.

4.7.5 Burst pressure. The cylinder shall be hydrostatically pressurized, at a rate not greater than 25,000 psig (172,650 kPa) per minute to a maximum pressure of 5000 psig (34,470 kPa) and the pressure maintained for three minutes. The pressure shall then be gradually increased to destruction and the burst pressure recorded. The cylinder shall conform to the requirements of 3.5.5.

4.7.6 Endurance test. The cylinder shall be hydrostatically charged to 2100 +/- 50 psig (14,500 +/-345 kPa) and the pressure released to obtain an internal pressure of 50 +/- 50 psig (345 +/-345 kPa). This shall constitute one cycle. The charging and discharging shall be repeated for 20,000 cycles. The cylinder shall pass the requirements specified in 3.5.6. The cylinder shall then be subjected to and pass the permanent volumetric expansion test.

4.7.7 Flattening. The cylinder shall be flattened to six times the wall thickness between wedge-shaped knife edges with a 60-degree (1.05 rad) included angle and rounded to a 1/2 inch (13 mm) radius. The knife edge shall be applied perpendicularly to the longitudinal axis of the cylinder at the center of the cylinder. For longitudinal welds, the bend knuckle shall occur on the weld. The cylinder shall pass the requirements specified in 3.5.7.

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4.7.8 Macrostructure (deep acid etch). Longitudinal sections from the cylinder shall be etched in an aqueous solution containing 50 percent hydrochloric acid by volume and maintained at a temperature of 160 +/- 10 degrees F (71 degrees C) for sufficient time to show the presence of defects. The sections shall then be visually examined by a metallographist and shall pass the requirements specified in 3.5.8.

4.7.9 Radiographic inspection. The welded sections of cylinders, except brackets welded to the spiral cylinder, shall be inspected in accordance with MIL-STD-453. The welded sections shall pass the requirements specified in 3.5.9.

4.7.10 Fragmentation resistance. Oxygen, conforming to MIL-O-27210, Type I (see 4.6.1), shall be used in the performance of this test.

4.7.10.1 Procedure I. Cylinders greater than 2-1/2 inches in diameter shall be subjected to gunfire under the following conditions:

a. The cylinder shall be charged to 2100 +/- 50 psig (14,500 +/- 345 kPa) at gun range ambient air temperature.

b. The cylinder may be supported but not constrained.

c. The ammunition shall be .50 caliber M-2, armor-piercing.

d. The range shall be 50 yards (45.7 m) maximum.

e. The various cylinders taken for test shall be tested, each progressively, in a different position. The point of impact shall be near the weld. The cylinder positions shall be as follows:

(1) With the longitudinal axis of the cylinder normal to the line of fire.

(2) With the longitudinal axis of the cylinder 45 degrees (.785 rad) from normal toward the gun position.

(3) With the longitudinal axis of the cylinder parallel to the line of fire with inlet port away from the gun position.

f. All shots shall be tumbled. A suggested method for projectile tumbling is shown on Figure 1.

g. The tumbled projectile shall have a velocity of not less than 2,600 ft/sec (792 m/s) at the point of impact with the cylinder.

h. To verify that the projectile was satisfactorily tumbled, the size entry hole made by the projectile shall be not less than one-half inch by one and one-half inch (13 mm by 39 mm).

i. Verify that the projectile trajectory and tumble are satisfactory. This may be determined by the location and visual appearance of a hole made by the passage of the projectile through vertically suspended sheets of paper at the target area.

The cylinder shall pass the requirements specified in 3.5.10.1.

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4.7.10.2 Procedure II. Cylinders 2-1/2 inches or less in diameter shall be subjected to gunfire under the following conditions:

a. The cylinder shall be charged to 2100 +/-50 psig (14,500 +/- 345 kPa) at gun range ambient air temperature.

b. The cylinder may be supported but not constrained.

c. The cylinder shall be positioned as indicated in 4.7.10.1e.

d. The ammunition shall be .30 caliber armor piercing, with velocity of 2800 +/- 100 feet per second (854 +/- 30 m/s).

e. The range shall be approximately 20 yards (18.3 m).

f. Shots shall not be tumbled.

The cylinders shall pass the requirements specified in 3.5.10.2.

4.7.11 Physical properties. Samples of materials from cylinders specified in 4.3.1 shall be tested for tensile strength and elongation in accordance with ASTM E 8. The samples shall pass the requirements specified in 3.5.11.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or C, as specified in the contract (see 6.2.1j).

5.1.1 Level A. Each cylinder shall be preserved and packaged in accordance with Method IC of MIL-P-116, without the use of contact preservatives. Cleaning shall be accomplished without the use of petroleum derivative solvents. The cylinder port shall be sealed with a metal cap or plug conforming to MIL-C-5501 or commercially equal metal cap or plug. When installing aluminum caps or plugs, caution shall be exercised in the application and tensioning of same to preclude the shearing of particles from the softer metal fittings. After cleaning, package each cylinder in a close-fitting box conforming to PPP-B-636. The container closure shall be accomplished in accordacne with applicable procedures specified in the appendix of PPP-B-636.

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5.1.2 Level C. The cylinder shall be individually packaged in a manner that will prevent physical damage or environmental deterioration during transit from the shipping point to the original consignee. As a minimum, the cylinder port shall be plugged or capped, as applicable, with metal fittings only, conforming to MIL-C-5501, or commercial equal. When installing aluminum caps or plugs, caution shall be exercised in the application and tensioning of same to preclude shearing particles off the softer metal fittings.

5.2 Packing. Packing shall be Level A, B or C, as specified by the contract (see 6.2.1j). Insofar as practicable, shipping containers shall effect a close fit, contain identical quantities and be of uniform dimensional configuration. For Levels A and B, the gross weight of boxes shall be not greater than 200 pounds (90.7 kg).

5.2.1 Level A. A cylinder packaged as specified in 5.1.1 shall be packed for shipment in a Fiberboard Corrugated, Triple Wall box conforming to PPP-B-640, Class 2 (Weather Resistant), Style E. The closure sealing and reinforcing (banding) of the container shall be accomplished in accordance with the procedures specified in the appendix of the container specification.

5.2.2 Level B. A cylinder packaged as specified in 5.1.1 shall be packed for shipment as specified in 5.2.1, except that the fiberboard shipping container conforming to PPP-B-640 shall be Class 1 (Nonweather Resistant), Style E.

5.2.3 Level C. A cylinder packaged as specified in 5.1.1 shall be packed for shipment in a manner that will afford protection against damage during transit from the shipping point to the original consignee. Containers used shall be in accordance with Uniform Freight Classification Rules, or regulations of other carriers applicable to the mode of transportation utilized.

5.3 Marking. Each unit and shipping container shall be marked in accordance with MIL-STD-129. In addition, the following or similar precautionary marking shall be required on each unit package and shipping container.

CAUTION

DO NOT ALLOW PETROLEUM CONTAMINANTS OF ANY KIND
TO BE USED/STORED ON OR ABOUT THESE CYLINDERS.

6. NOTES

6.1 Intended use. The cylinders covered by this specification are intended for use in storing aviator's breathing oxygen.

6.2 Ordering data.

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6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification, including amendments/ revisions.

b. Applicable Qualified Products List (QPL), including date.

c. Government Part Number (see 3.4).

d. National Stock Number.

e. Applicable Government drawings, including revisions (see 4.4.1).

f. Quantity desired.

g. Whether first article inspection is waived (see 4.4.1).

h. The name and address of the first article inspection laboratory (see 4.4.1) and the name of the Government activity responsible for conducting the first article inspection program (see 4.4.1.1).

i. Cylinders subjected to destructive tests (see Table III) should not be shipped as part of the acquisition quantity.

j. Applicable levels of preservation, packaging, and packing (see 5.1 and 5.2).

k. Marking requirements (see 5.3).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs:

Paragraph No.	Data requirement title	Applicable DID no.	Option
4.4.1.1	Inspection and Test Reports	DI-T-5329	----

(Data item descriptions related to this specification, and identified in Section 6 will be approved and listed as such in DOD 5010.12-L, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

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6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List (QPL) whether or not such products have actually been so listed by that date. The attention of contractors is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Commander, Naval Air Systems Command, Department of the Navy, Washington, D.C., 20360; however, authorization for qualification of products shall be obtained from the Commanding Officer, Naval Air Development Center, Warminster, PA, 18974-5000, Attention: Code 6031.

6.3.1 Drawings. When requested, the manufacturer should submit engineering drawings and inspection reports in accordance with SD-6.

6.4 First article. When a first article is required, the item should be tested and should be a first article sample. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for examinations, tests and approval of the first article.

6.5 Subject term (key word) listing.

Burst pressure
Cleanliness
Closure
Color
Cylinders, oxygen
Endurance
External threads
Finish
First article
Flattening
Fragmentation resistance
Identification
Internal threads
Leakage
Macrostructure defects
Neck
Permanent volumetric expansion
Physical properties
Qualification
Quality conformance
Radiographic inspection
Spiral cylinder
Steel
Vibration
Weight
Welding
Workmanship

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6.6 International standardization. Certain provisions (3.9.2), of this specification are the subject of international standardization agreement (STANAG 3056). When amendment, revision, or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization offices to change the agreement or make other appropriate accommodations.

Custodians:
Army - AV
Air Force - 99

Review activities:
Air Force - 11 and 71
International interest (see 6.6)

Preparing activity:
Navy - AS

(Project 8120-1003)

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Table I. Qualification inspection

Inspection	Requirement Paragraph(s)	Test method Paragraph(s)
Visual examination	3.5.1.1	4.7.1.1
Dimensional check	3.5.1.2	4.7.1.2
Cleanliness	3.5.2.1, 3.5.2.2, 3.5.2.3	4.7.2.1, 4.7.2.2, 4.7.2.3
Leakage	3.5.3	4.7.3
Permanent volumetric expansion	3.5.4	4.7.4
Burst pressure	3.5.5	4.7.5
Endurance	3.5.6	4.7.6
Flattening	3.5.7	4.7.7
Macrostructure defects	3.5.8	4.7.8
Radiographic inspection	3.5.9	4.7.9
Fragmentation resistance	3.5.10.1 or 3.5.10.2, as applicable	4.7.10.1 or 4.7.10.2, as applicable
Vibration (spiral cylinder)	3.5.11	4.7.11
Physical properties	3.5.12	4.7.12

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Table II. First article inspection

Inspection	Requirements Paragraph(s)	Test method Paragraph(s)
Visual examination	3.5.1.1	4.7.1.1
Dimensional check	3.5.1.2	4.7.1.2
Cleanliness	3.5.2.1, 3.5.2.2, 3.5.2.3	4.7.2.1, 4.7.2.2, 4.7.2.3
Leakage	3.5.3	4.7.3
Permanent volumetric expansion	3.5.4	4.7.4
Burst pressure	3.5.5	4.7.5
Flattening	3.5.7	4.7.7
Radiographic inspection	3.5.9	4.7.9

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Table III. Quality conformance inspection

Inspection	Test Method Paragraph(s)	Sample size	Acceptance criteria
Visual examination	4.7.1.1	a. Every cylinder for critical defects. b. Inspection Level II for minor defects.	a. Reject all units with any critical defect. b. An acceptable quality level of 2.5 defects per hundred units for minor defects.
Dimensional check	4.7.1.2	Inspection Level S-2 1/	Acceptance number zero, rejection number 1.
Cleanliness	4.7.2.1, 4.7.2.2, 4.7.2.3	Inspection Level S-3 1/	Acceptance number zero, rejection number 1.
Leakage	4.7.3	Every cylinder	Reject all defective units.
Permanent volumetric expansion	4.7.4	Every cylinder	Reject all defective units.
Burst pressure 2/	4.7.5	Inspection Level S-2 1/	Acceptance number zero, rejection number 1.
Flattening 2/	4.7.7	Inspection Level S-1 1/	Acceptance number zero, rejection number 1.
Radiographic inspection	4.7.9	Every cylinder	Reject all defective units.
Physical properties 2/	4.7.12	Inspection Level S-1 1/	Acceptance number zero, rejection number 1.
Packaging	4.7.1.3	Inspection Level S-2	Total acceptable quality level of 4.0 percent defective.

1/ The sample size shall be based only on the applicable sample size code letter corresponding to the specified inspection level of MIL-STD-105.

2/ This is a destructive test (see 6.2.11).

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Table IV. Classification of defects for visual examination of the cylinder

Defects	Critical	Minor
Incorrect outlet configuration	X	
Incorrect or defective threads	X	
Incorrect, incomplete, or faulty finish		X
Incorrect part number or cylinder code number	X	
Flaw in material	X	
Incorrect or incomplete marking		X
Incorrect color		X
Abnormally distorted or nonuniform weld	X	
Loose particles inside the cylinder	X	
Rust or scale	X	
Irregularities in the internal surface greater than 1/16 inch.	X	
Existence of any foreign material, solid or liquid.	X	

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Table V. List of defects for packaging

Item	Defects
Exterior and interior markings	Missing, incorrect, incomplete, illegible; of improper size, location, sequence, or method of application; markings not the same on the interior and exterior containers.
Packaging and packing materials	Any nonconforming component; any component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of the components such as incomplete closure of the unit package, intermediate package, container flaps, loose strappings, etc.; bulging or distortion of the containers.
Exterior and interior weight or content.	Number per container is more or less than required; gross or net weight exceeds the requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-29576	Cylinders, Oxygen, Steel Nonshatterable, Welded, 2100 PSI

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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Naval Air Engineering Center
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